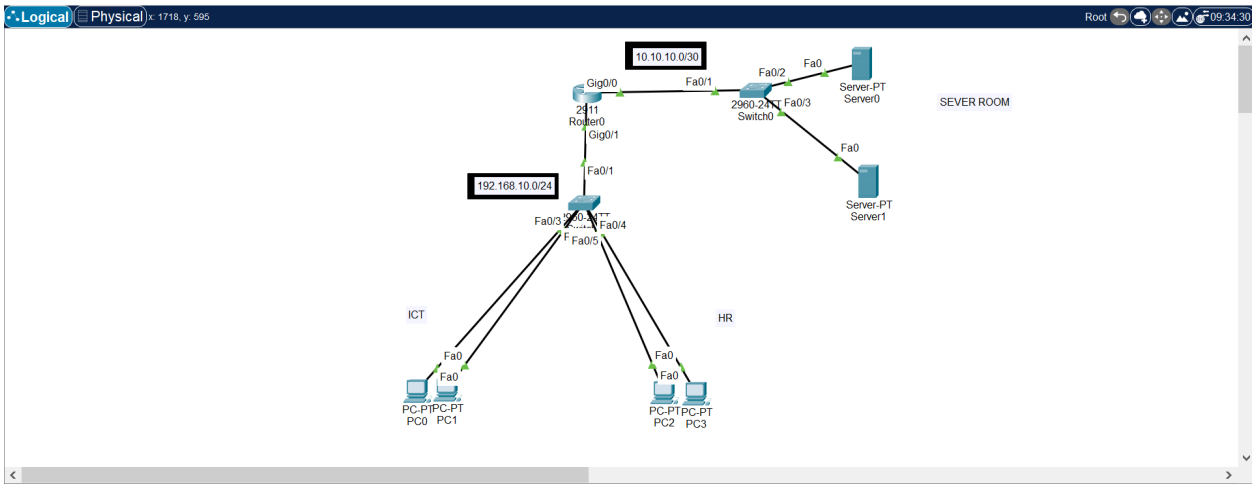
DESIGN AND IMPLEMENTATION OF ACCESS CONTROL LIST (ACL) FOR NETWORK SECURITY.

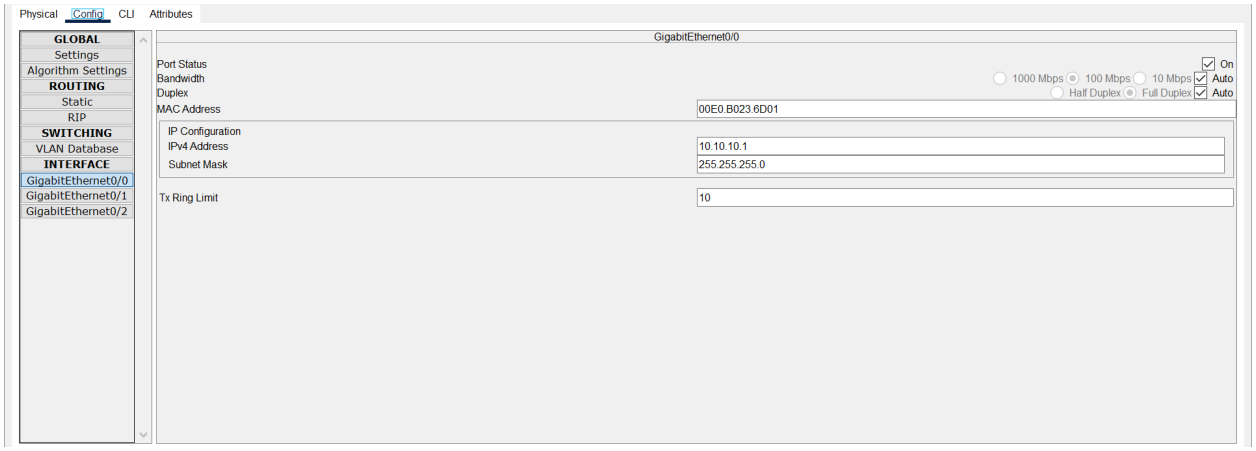
Network topology:

Logical topology showing 1 router, 2 switches, 4PCs [2PCs for department] and 2 servers. The router serves as the gateway interconnecting both department and enforcing access control



Router interface IP assignment:

IP addresses were assigned to routers interfaces using the config tab in packet tracer. Each interface was selected, IP and subnet mask entered, port status turned on.

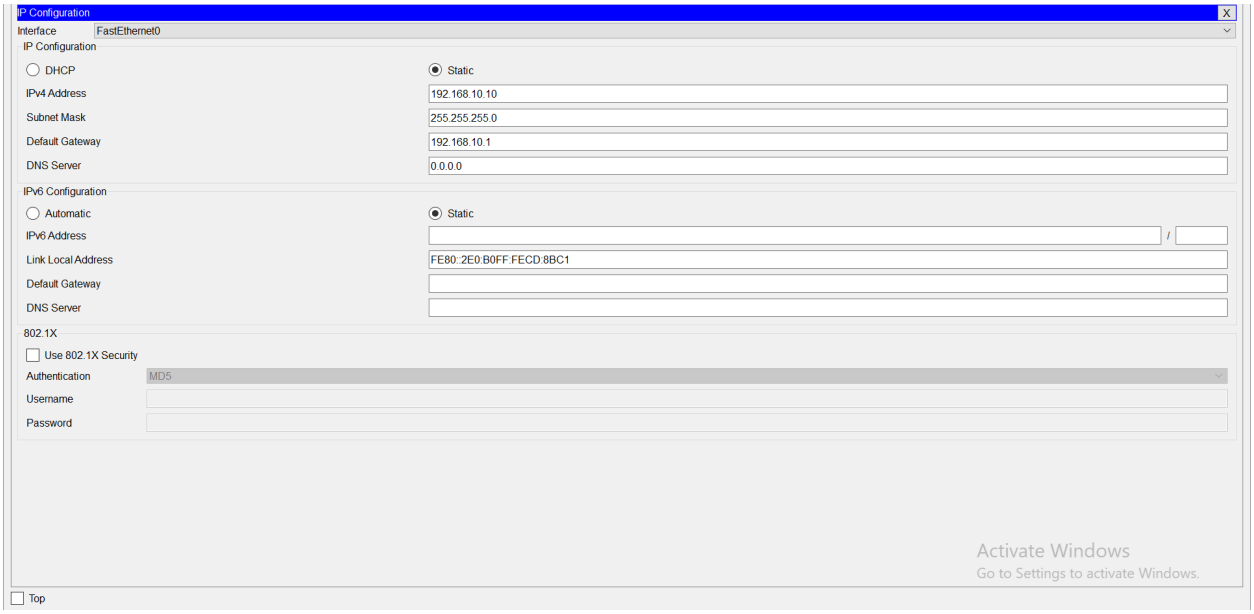


Router details used :

1) FastEthernet0/1= 192.168.10.1 255.255.255.0- LAN gateway for both ICT & HR
2) FastEthernet0/0= 10.10.10.1 255.255.255.252 - WAN

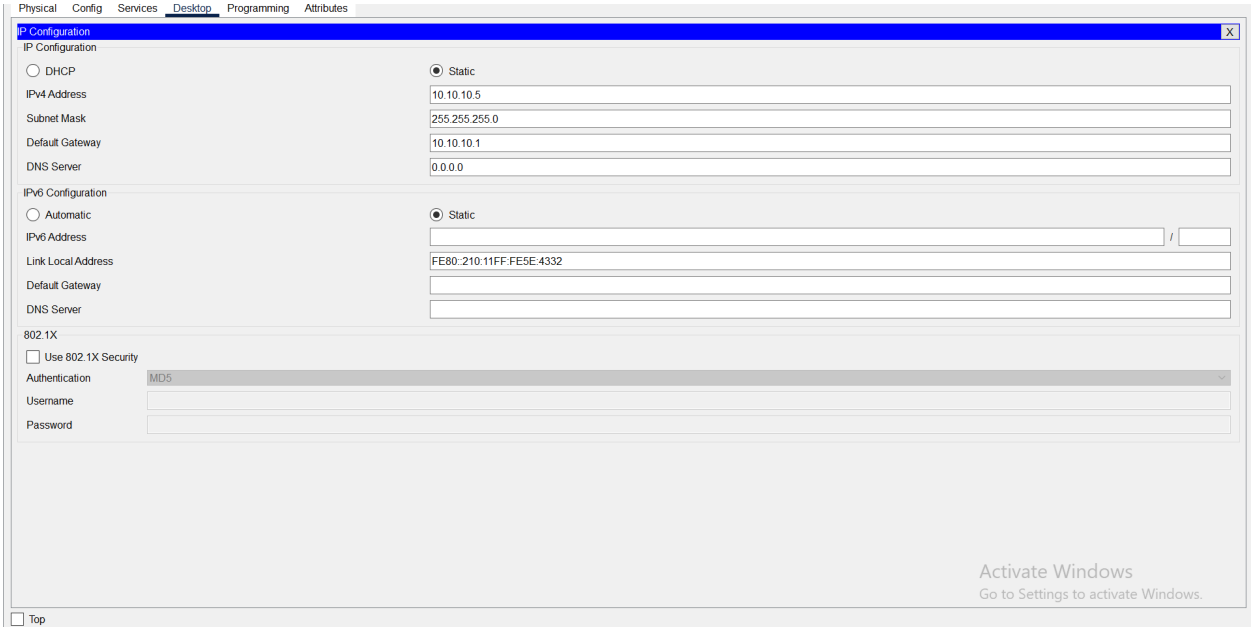
PCs IP Assignment:

The 4 PCs were assigned IP addresses statically via desktop > IP configuration. All PCs use a subnet mask of 255.255.255.0 with the gateway as 192.168.10.1. ICT PCs and HR PCs were placed on the same network but in different departments using a single switch.



Server IP Assignment :

Servers were assigned IP addresses statically within the same network 10.10.10.0 to allow centralised access with the gateway of 10.10.10.1.



ACL Configuration :

Access Control List [ACL] was configured on the router to enforce departmental security policy. A standard ACL was used to filter traffic based on source IP and destination IP.
The ACL denies HR access to servers while allowing ICT full access to servers.

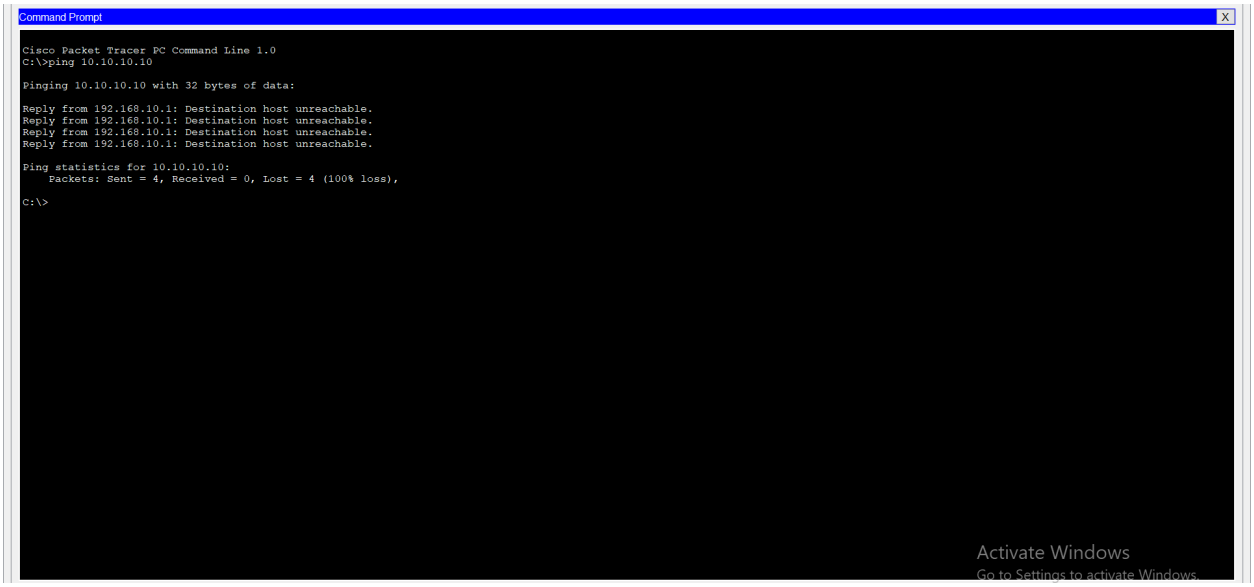
```
Router(config-if)#access-list 10 permit host 192.168.10.11
Router(config)#access-list 10 permit host 192.168.10.11
Router(config)#access-list 10 permit host 192.168.10.10
Router(config)#access-list 10 deny any
Router(config)#int gig0/1
Router(config-if)#ip access-group 10 in
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Testing and Results:

Successful access test from ICT department. PC with IP 192.168.10.10 pinging server 10.10.10.5 shows reply, confirming ICT is allowed access to servers as per policy.



Denied access test from HR department. PC with IP 192.168.10.100 pinging server 10.10.10.10 shows destination unreachable, confirming ACL is blocking HR access to servers.



Conclusion:

The ACL implementation successfully achieved departmental isolation. The HR department was denied access to critical servers while the ICT department was allowed to retain full access, demonstrating effective network security control using standard ACL on the router.